

# MEDTRACK\_DV

## Hospital Operations & Patient Analytics Dashboard

Project Documentation

Infosys Springboard 7.0 – Data Visualization (Batch -2)

Implementation using Power BI and Tableau

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## 1. Project Overview

MedTrack\_DV is a hospital operations and patient analytics project developed as part of Infosys Springboard 7.0. The project takes hospital data through data preparation, KPI generation and dashboard development to present useful information through Power BI and Tableau.

The project is organized into eight modules so that each stage has a clear input and output.

| Module   | Work                                          |
|----------|-----------------------------------------------|
| Module 1 | Data Collection / Generation and Integration  |
| Module 2 | Data Cleaning and Preparation                 |
| Module 3 | Hospital KPI Engineering                      |
| Module 4 | Dashboard Planning and Storyboard             |
| Module 5 | Hospital Overview and Patient Flow            |
| Module 6 | Department Analytics and Resource Utilization |
| Module 7 | Final Integration, Testing and Validation     |
| Module 8 | Documentation and Project Delivery            |

### Four Dashboard Areas

| Dashboard            | Main purpose                                        |
|----------------------|-----------------------------------------------------|
| Hospital Overview    | Overall hospital performance and major KPIs         |
| Patient Flow         | Admissions, patient stay and readmission analysis   |
| Department Analytics | Department-level volume, performance and efficiency |
| Resource Utilization | Bed, ward and patient-load analysis                 |

## 2. Problem Statement and Objectives

Hospitals generate information from patient admissions, department activity, bed status and other operational activities. When these measures are viewed separately, it can be difficult to understand the overall operational situation.

MedTrack\_DV brings the available measures into an analytical environment so that hospital information can be viewed through KPIs, charts, filters and dashboard interactions.

### Objectives

- Prepare a structured hospital dataset for analysis.
- Clean and standardize the available data.
- Calculate important hospital KPIs.
- Plan the dashboard layout before development.
- Develop four dashboards covering major hospital operations.
- Use Power BI and Tableau for dashboard implementation.
- Test calculations, filters and dashboard behavior.
- Organize and document the final project for GitHub delivery.

## 3. Tools and Technologies

| Tool / Technology | Use in the project                                     |
|-------------------|--------------------------------------------------------|
| Python            | Data preparation and KPI-related processing            |
| Pandas / NumPy    | Data cleaning, transformation and numerical processing |
| Jupyter Notebook  | Hospital data-cleaning workflow                        |

|          |                                                |
|----------|------------------------------------------------|
| CSV      | Hospital raw/cleaned datasets                  |
| Excel    | Final analytical dataset and KPI summaries     |
| Power BI | Dashboard development                          |
| Tableau  | Dashboard development                          |
| GitHub   | Project organization and delivery              |
| PDF      | Dashboard storyboard and project documentation |

## 4. Module 1 – Data Collection / Generation & Integration

Module 1 establishes the data foundation for the project. The hospital information is collected or prepared and arranged into a raw dataset for the cleaning stage.

- Prepare the required hospital and patient admission information.
- Collect or generate the required data fields.
- Integrate the information into the raw hospital dataset.
- Use the raw dataset as the input for Module 2.

| File                  | Purpose                                                         |
|-----------------------|-----------------------------------------------------------------|
| data_collection.py    | Python script used for the data collection / preparation stage. |
| hospital_raw_data.csv | Raw hospital dataset used as the source for cleaning.           |

## Module 2 – Data Cleaning & Preparation

The cleaning stage converts the raw data into a consistent dataset suitable for KPI generation and dashboard analysis. The submitted Jupyter Notebook contains the cleaning workflow.

- Check missing and duplicate values.
- Standardize column names and categorical values.
- Standardize department and gender information.
- Convert and validate date fields.
- Validate patient age and create useful age-related fields.
- Prepare time-related fields for later analysis.
- Save the cleaned dataset as a separate CSV file.

| File                    | Purpose                                          |
|-------------------------|--------------------------------------------------|
| hospital_cleaning.ipynb | Notebook used for data cleaning and preparation. |
| hospital_cleaned.csv    | Cleaned dataset used by the later modules.       |

## 5. Module 3 – Hospital KPI Engineering

Module 3 uses the cleaned dataset to calculate the hospital KPIs needed for dashboard analysis. The KPI-generation script also creates additional analytical fields and summary sheets.

| File                        | Purpose                                                             |
|-----------------------------|---------------------------------------------------------------------|
| generate_hospital_kpis.py   | Python script used for hospital KPI generation.                     |
| hospital_final_dataset.xlsx | Final analytical workbook containing the dataset and KPI summaries. |

## KPIs generated

| KPI                         | Meaning                                          |
|-----------------------------|--------------------------------------------------|
| Total Admissions            | Total number of admission records.               |
| Occupancy Rate              | Hospital occupancy indicator.                    |
| Average Length of Stay      | Average duration of patient stay.                |
| Readmission Rate            | Percentage of patients identified as readmitted. |
| Bed Utilization Rate        | Bed-use indicator based on the prepared data.    |
| Department Efficiency Score | Relative department performance score.           |

## KPI workbook sheets

- Final\_Dataset
- KPI\_Summary
- KPI\_Definitions
- Department\_Analytics
- Monthly\_KPIs

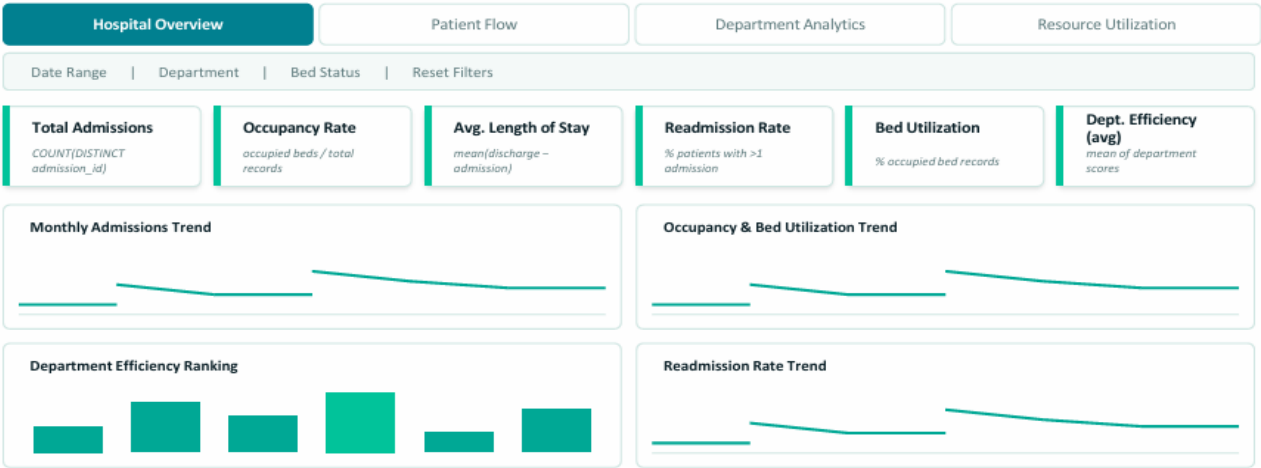
## 6. Module 4 – Dashboard Planning & Storyboard

Module 4 is the planning stage for the dashboard development. The storyboard is used to decide the structure, information placement and analytical flow before final dashboard work.

| Dashboard            | Planned focus                                        |
|----------------------|------------------------------------------------------|
| Hospital Overview    | Hospital KPIs, admissions and overall performance    |
| Patient Flow         | Admissions, stay-related measures and readmission    |
| Department Analytics | Department volume, LOS and efficiency comparison     |
| Resource Utilization | Beds, ward types, patient load and resource analysis |

Deliverable: dashboard\_Storyboard.pdf

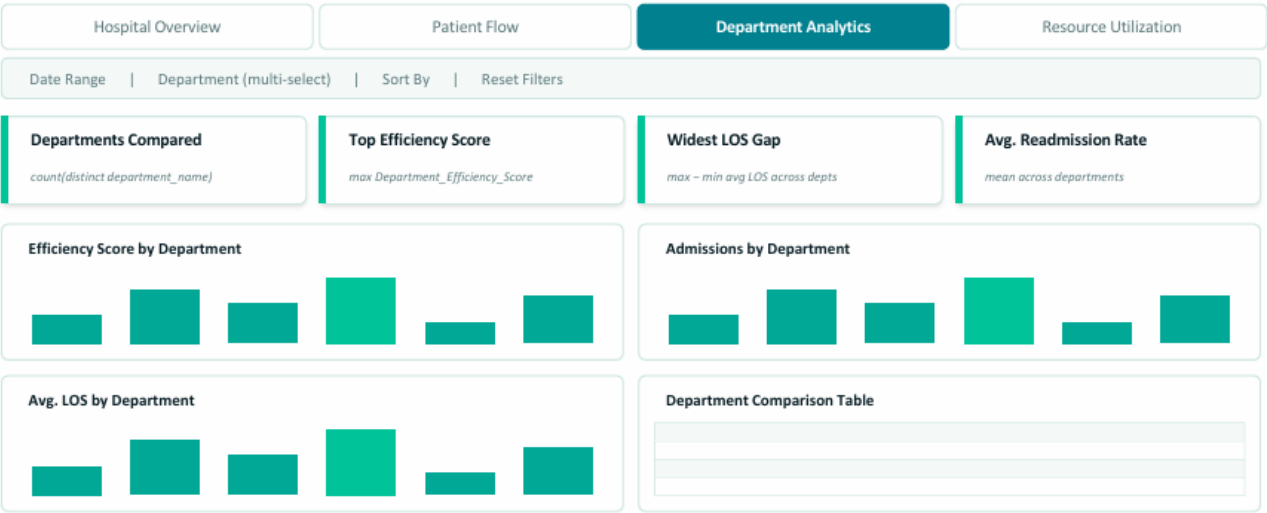
# Hospital Overview



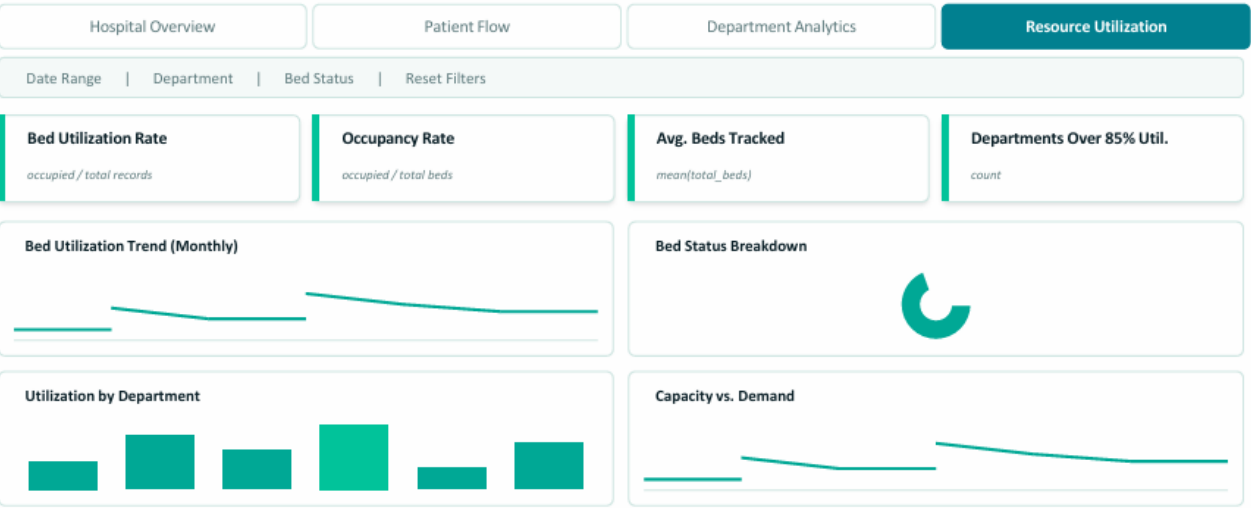
# Patient Flow



## Department Analytics



## Resource Utilization



## 7. Module 5 – Hospital Overview & Patient Flow

Module 5 covers the first two dashboard areas. The dashboard development version in the repository is `medtrack_dashboard_v1.pbix`.

### Hospital Overview

- Main hospital KPIs.
- Admission type mix.
- Patient distribution by department.
- Admission analysis by year.
- Disease-category analysis.
- Hospital-level operational measures.

### Patient Flow

- Admission trends by month.
- Length-of-stay distribution.
- Patient stay distribution.
- Readmission-status analysis.

## 8. Module 6 – Department Analytics & Resource Utilization

Module 6 covers the remaining two dashboard areas and completes the planned four-dashboard scope.

### Department Analytics

- Average length of stay by department.
- Admission count by department.
- Department performance comparison.
- Admission by gender.
- Readmission trend by year.

### Resource Utilization

- Department contribution to hospital load.
- Patient load compared with length of stay.
- Length of stay by admission type.
- Bed status by ward type.
- Patient load by ward type.

## 9. Module 7 – Final Integration, Testing & Validation

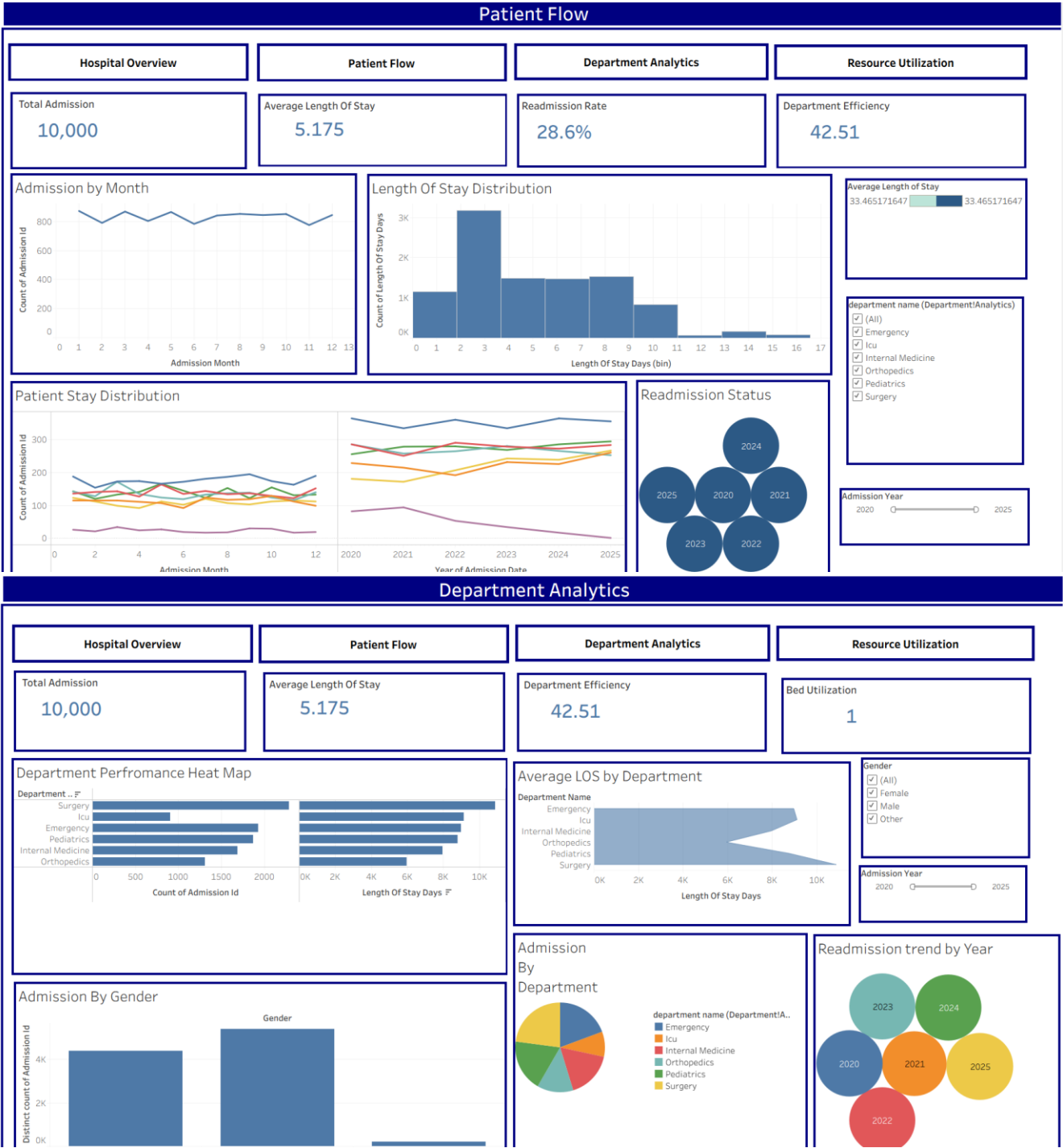
Module 7 contains the final integrated dashboard files and the final validation work.

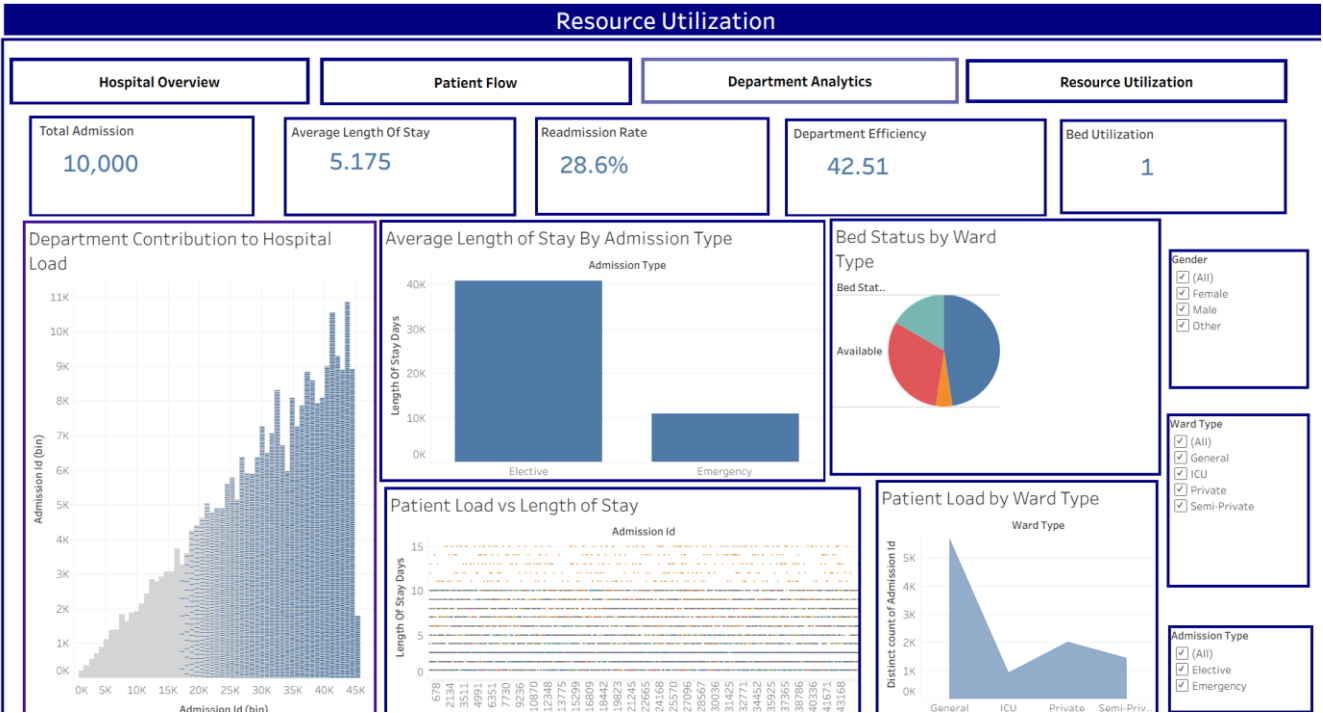
| Final file       | Purpose                                                   |
|------------------|-----------------------------------------------------------|
| MedTrack_DV.pbix | Final Power BI report.                                    |
| MedTrack_DV.twbx | Final Tableau workbook containing the dashboard solution. |

### Testing checklist

| Test area         | Check                                                  |
|-------------------|--------------------------------------------------------|
| KPI calculations  | Values agree with the prepared analytical dataset.     |
| Data consistency  | Correct dataset and fields are used.                   |
| Filters           | Filters affect the intended visuals and KPIs.          |
| Navigation        | Navigation works where implemented.                    |
| Dashboard actions | Interactive selections behave correctly.               |
| Power BI          | Final PBIX opens and visuals/interactions work.        |
| Tableau           | Final TWBX opens and dashboards/interactions work.     |
| Final review      | Labels, layout, readability and usability are checked. |







## 10. Module 8 – Documentation & Project Delivery

Module 8 organizes the project for final submission and GitHub delivery. Documentation should make it easy to understand the work without opening every project file.

### Project identification

| Item          | Details                                                         |
|---------------|-----------------------------------------------------------------|
| Name          | Pricilla G                                                      |
| College       | Arunachala College of Engineering for Women                     |
| Course        | B.Tech – Artificial Intelligence & Data Science                 |
| GitHub Branch | Pricilla-G                                                      |
| Project       | MedTrack_DV – Hospital Operations & Patient Analytics Dashboard |

### Recommended Module 8 contents

- README.md
- Final project documentation
- Milestone documentation
- Final submission notes, if required by the mentor

## 11. Recommended GitHub Folder Structure

Based on the files shown in the repository screenshot, the following arrangement is cleaner. It avoids keeping all files in the root and avoids duplicating final dashboard files.

| Folder   | Files                                                    |
|----------|----------------------------------------------------------|
| MODULE 1 | data_collection.py<br>hospital_raw_data.csv              |
| MODULE 2 | hospital_cleaning.ipynb<br>hospital_cleaned.csv          |
| MODULE 3 | generate_hospital_kpis.py<br>hospital_final_dataset.xlsx |
| MODULE 4 | dashboard_Storyboard.pdf                                 |
| MODULE 5 | medtrack_dashboard_v1.pbix                               |
| MODULE 6 | Module 6 dashboard files/screenshots, if available       |
| MODULE 7 | MedTrack_DV.pbix<br>MedTrack_DV.twbx                     |
| MODULE 8 | Final documentation<br>Milestone documentation           |
| ROOT     | README.md                                                |

## 12. Results and Conclusion

### Project results

- The cleaned dataset contains 10,000 hospital records.
- Six hospital KPIs were generated.
- The final analytical workbook contains KPI and department/monthly analysis sheets.
- Four dashboard areas were developed.
- Power BI and Tableau files are included in the project.

- The project follows a complete workflow from data preparation to dashboard delivery.

### Selected observations from the submitted data

- Surgery has the highest number of admissions, with 2,291 admissions.
- ICU has the highest average length of stay, at approximately 10.08 days.
- Orthopedics has the highest department readmission rate in the submitted department summary, at approximately 30.05%.
- The highest monthly admission count in the submitted monthly data is 162 in March 2025.
- The highest monthly readmission rate in the submitted monthly data is 36.36% in March 2023.

### Note

The final KPI workbook reports Occupancy Rate and Bed Utilization Rate as 0.00%. These values are retained as reported in the submitted workbook.

### Conclusion

MedTrack\_DV demonstrates how hospital data can be prepared, transformed into meaningful KPIs and presented through interactive dashboards. The project combines Python-based data processing with Power BI and Tableau to provide a clear view of hospital admissions, patient flow, department performance and resource utilization.